

Image Processing Tutorial

Spatial Filtering & Histogram Processing (Sheet 4 / Tutorial 4) • ECE 359

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Outline

- 1 1-D Spatial Filtering
- 2 2-D Spatial Filtering
- 3 Basic Filter Computations
- 4 Correlation vs Convolution
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Convolution & Correlation

1

1	2	3
0	8	6
7	8	9

$f * H$
 $\downarrow$ image $\downarrow$ kernel (Spatial Filter)

Correlation

$$H = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \rightarrow \begin{bmatrix} 6 & 8 & 4 \\ 9 & 5 & 7 \\ 3 & 2 & 1 \end{bmatrix}$$

Correlation $\rightarrow$ use H as it is
 Convolution $\rightarrow$ rotation 180°

To prevent confusion H Matrices used are for directly multiply to the selected portion of the image

Shift filters

Image Filter

$$\begin{bmatrix} 10 & 50 & 60 \\ 80 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 8$$

Identity matrix

Correlation based matrices

$$\begin{bmatrix} 0 & 0 & 0 \\ 10 & 50 & 60 \\ 80 & 8 & 9 \\ 10 & 11 & 12 \\ 0 & 0 & 0 \end{bmatrix}$$

Shift left

H

$$\begin{bmatrix} 10 & 50 & 60 \\ 80 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = 9$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 10 & 50 \\ 0 & 80 & 8 \end{bmatrix} \cdot \begin{bmatrix} 1 \end{bmatrix} = 50$$

Shift left Convolution

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{180^\circ} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Shift down

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

up

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Shift down

$$\begin{bmatrix} 10 & 50 & 60 \\ 80 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 50$$

Shift up

$$\begin{bmatrix} 10 & 50 & 60 \\ 80 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = 8$$

1-D Filtering — Average & Median (Problem 1)

Part (a)

Consider a row of pixels with values 1, 1, 1, 1, 5, 1, 1, 1, 1.

When we apply an **average** and a **median filter** of size 3, what would be the output values of the **5th pixel** from the left?

Part (b)

Consider a row of pixels with values 1, 1, 1, 1, 5, 5, 5, 5, 5.

When we apply an **average** and a **median filter** of size 3, what would be the output values of the **5th pixel** from the left?

Hint

Average filter: $\hat{p} = \frac{1}{3}(p_{i-1} + p_i + p_{i+1})$

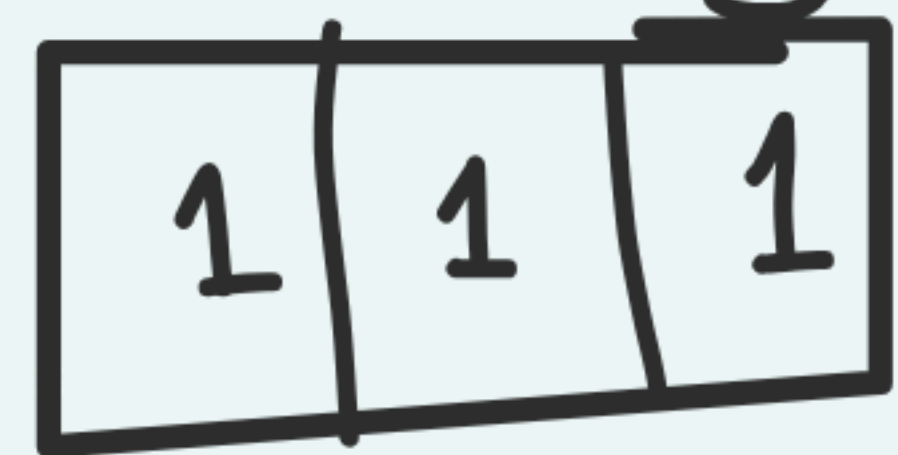
Median filter: $\hat{p} = \text{median}\{p_{i-1}, p_i, p_{i+1}\}$

Solution — Problem 1

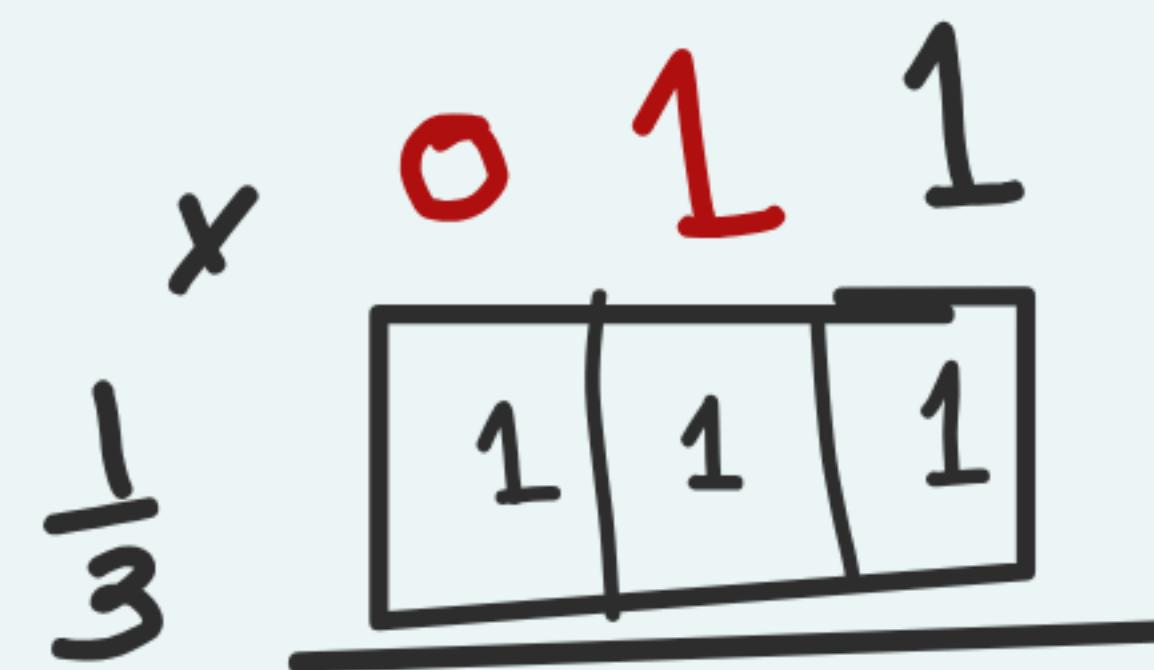
Problem 1 Solution

1, 1, 1, 1, 5, 1, 1, 1, 1

1D Averaging filter = $\frac{1}{N} (P_{i-1} + P_i + P_{i+1})$



$\frac{1}{3}$



$$= (0 + 1 + 1) \times \frac{1}{3} = \frac{2}{3} = 0.6 = 1$$

$$\rightarrow \hat{P}_5 = (1 \times 1 + 1 \times 5 + 1 \times 1) \frac{1}{3} = 2.3 \approx 2$$

Median

① Sorting

② choose middle number for center pixel.



$$\hat{P}_5 = \text{median}\{1, 5, 1\} = \text{median}\{1, \textcircled{1}, 5\} = 1$$

1, 1, 1, 1, 5, 5, 5, 5, 5

$$\textcircled{1} \hat{P}_{\text{avg}} = (1 \times 1 + 5 \times 1 + 5 \times 1) \frac{1}{3} = \frac{11}{3} = 2.6 \approx 3$$

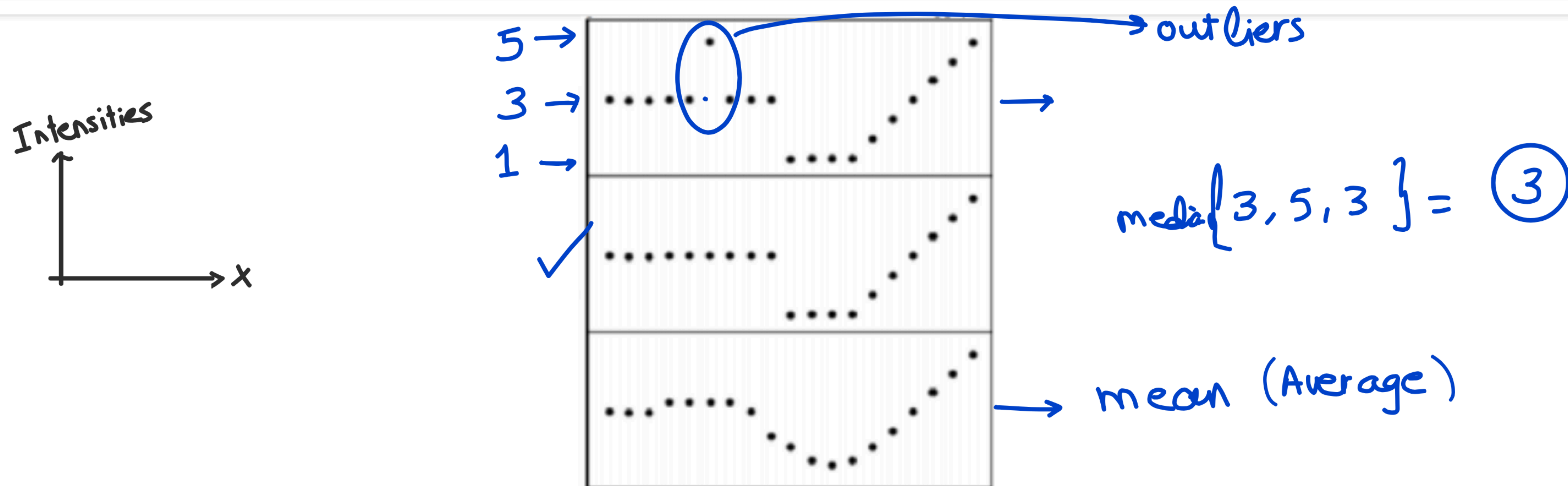
$$\textcircled{2} \hat{P}_{\text{med}} = 5$$

1-D Signal: Mean vs. Median Filter (Problem 2)

Problem 2

The **first** signal below is the original 1-D signal. The following two signals are the result of applying either a **median** or a **mean** filter.

State which filtered signal corresponds to each filter and give your reasoning.



Original signal (top), Filtered signal 1 (middle), Filtered signal 2 (bottom)

Reference Image (Used in Problems 3–6)

	0	1	2	3	4
0	100	102	104	102	106
1	102	105	107	106	106
2	104	48	108	109	110
3	105	110	104	106	109
4	107	105	103	107	105

5×5 image — indexed $[col, row]^T$ starting at $[0, 0]^T$

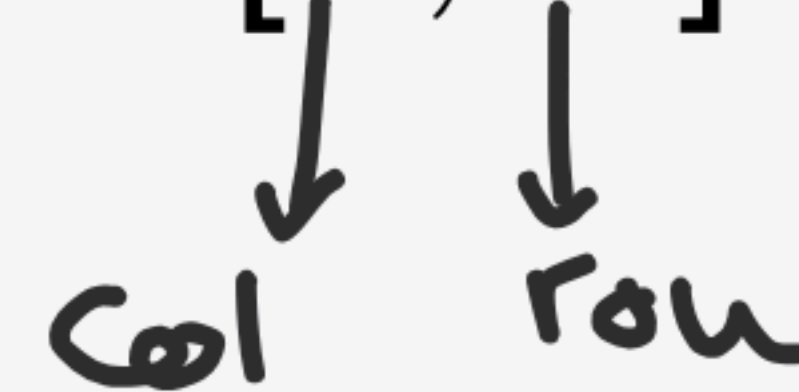
Indexing Convention

Coordinates are $[x, y]^T$ where $x =$ column, $y =$ row (0-based).

The value **48** at $[1, 2]^T$ is the outlier pixel.

3×3 Average & Median Filter (Problem 3)

Problem 3 — Average Filter

Using the reference image, what is the intensity at pixels $[2, 0]^T$ and $[3, 2]^T$ after a **3×3 average filter**?


Problem 3 (cont.) — Median Filter

What is the intensity at pixel $[2, 1]^T$ after a **3×3 median filter**?

Average kernel reminder

$$H = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Handle boundary pixels by **zero-padding** unless stated otherwise.

Solution — Problem 3

Problem 3 Solution

100	102	104	102	106
102	105	107	106	106
104	48	108	109	110
105	110	104	106	109
107	105	103	107	105

histogram

r_i	n
r_{109}	2
r_{110}	1
r_{106}	3

Average $\begin{matrix} \text{col} & \text{row} \\ [3, 2] \end{matrix}^T$

$$\hat{P}_{[3,2]} = 107.2 \approx 107$$

Problem 3

$$\hat{p}_{2,0} = \frac{1}{9} \cdot \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 102 & 104 & 102 \\ 105 & 107 & 106 \end{bmatrix} = 69.6 \approx 70$$

Problem 3 (b) median at $\begin{bmatrix} 2, 1 \end{bmatrix}$

48, 102, 102, 104, 105, 106, 107... $\hat{P}_2 = 105$
 $\downarrow$ median

- collect all values
- Sort
- choose
- replace

Gaussian Filter (Problem 4)

Problem 4

Using the reference image, what is the intensity at pixel $[2, 2]^T$ after applying the **Gaussian filter** with the weighting matrix below?

$$H = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Note: The sum of all kernel weights equals 1
($1+2+1+2+4+2+1+2+1 = 16$)

Problem 4 Solution

↓

100	102	104	102	106
102	105	107	106	106
→ 104	48	108	109	110
105	110	104	106	109
107	105	103	107	105

$$\hat{p}_5 = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 105 & 107 & 106 \\ 48 & 108 & 109 \\ 110 & 104 & 106 \end{bmatrix} = 100$$

$$= 1(105) + 2(107) + 1(106) + 2(48) + 4(108) + 2(109) + 1(110) + 2(104) + 1(106) = 99.7 \approx 100$$

Reverse-Engineering a Linear Filter (Problem 5)

Problem 5

After passing the reference image through an **unknown linear filter**, the result is shown below. Can you **identify the filter matrix** and state **how boundary pixels were handled**?

100	102	104	102	106
102	105	107	106	106
104	48	108	109	110
105	110	104	106	109
107	105	103	107	105

0	0	0	0	0
0	100	102	104	102
0	102	105	107	106
0	104	48	108	109
0	105	110	104	106

Filtered output image — Problem 5

Zero padding

Shift right → Shift down

$a \times b$
 $b \times a$

Blurring + Median on Binary Image (Problem 6)

Problem 6

Image $b(x, y)$ is binary (equal numbers of 0 and 255 pixels). Two filters are applied sequentially:

- **3×3 blurring filter** with coefficients $1/5$: $H = \begin{bmatrix} 0 & \frac{1}{5} & 0 \\ \frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ 0 & \frac{1}{5} & 0 \end{bmatrix}$
- **3×3 Median filter**

(a)

If blurring is applied **before** median, what are the **possible output pixel values** of $o(x, y)$?

and viceversa

(b)

Give a binary image $b(x, y)$ for which $o(x, y)$ is the **same regardless of filter order**.

Problem 6 Solution

Apply blurring filter

$$f(x,y) = \begin{bmatrix} 0 & 0 & 0 \\ 255 & 255 & 255 \\ 255 & 0 & 255 \end{bmatrix} \rightarrow$$

K: number of '255' pixels Multiplied

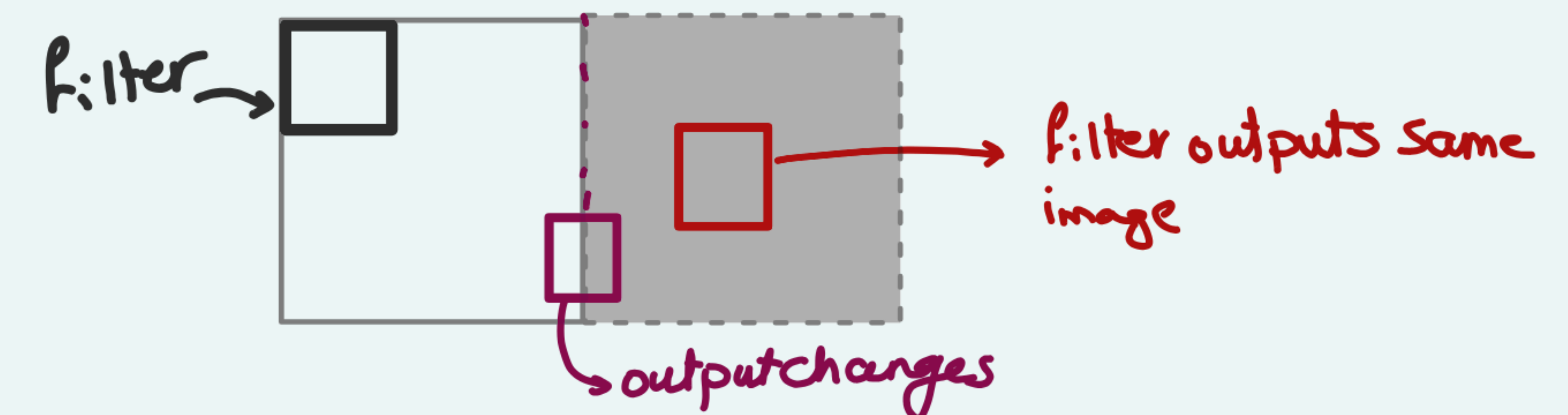
K	Blurred value
5	$255 \times \frac{5}{5} = 255$
4	204
3	153
2	102
1	51
0	0

possible outputs { 0, 51, 102, 153, 204, 255 }

⑥ Any image with pixels uniformly distributed as zeros or ones.
Hence, by applying blurring or median image doesn't change

→ All 0's $\Rightarrow$ 3x3 neighborhood
 ↳ Average of all 0's is a zero
 ↳ same goes for median

ex: half black half white image
and filter applied far from boundary.



Computing Filter Output (Problem 7)

Given: A 3×3 image region and averaging filter ✓

Image Region

$$\begin{bmatrix} 10 & 20 & 30 \\ 40 & 50 & 60 \\ 70 & 80 & 90 \end{bmatrix}$$

Filter

$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Find: Output value at center pixel

$$\hat{P}_5 = \frac{1}{9} [10 + 20 + \dots + 90] =$$

Median Filter Computation (Problem 8)

Given: A 3×3 neighborhood with values:

Neighborhood

$$\begin{bmatrix} 100 & 10 & 120 \\ 15 & 200 & 18 \\ 110 & 12 & 115 \end{bmatrix}$$

Tasks:

- Sort the values
- Determine the median for centre pixel $\hat{P}_3 = \text{median} \{10, 12, 15, 18, 100, 110, 115, 120, 200\} = 100$
- Compare with averaging filter result $\hat{P}_3 = \frac{1}{9} [10 + 12 + 15 + 18 + 100 + 110 + 115 + 120 + 200] =$

Note: Center pixel (200) appears corrupted by impulse noise